





Representation Learning for Recommender Systems

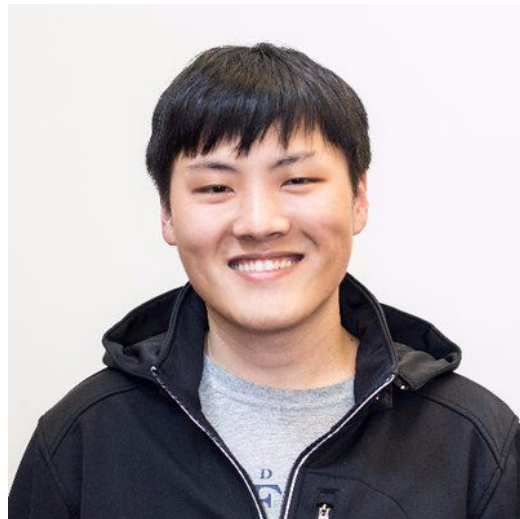
Andrew Zhai, Applied Scientist

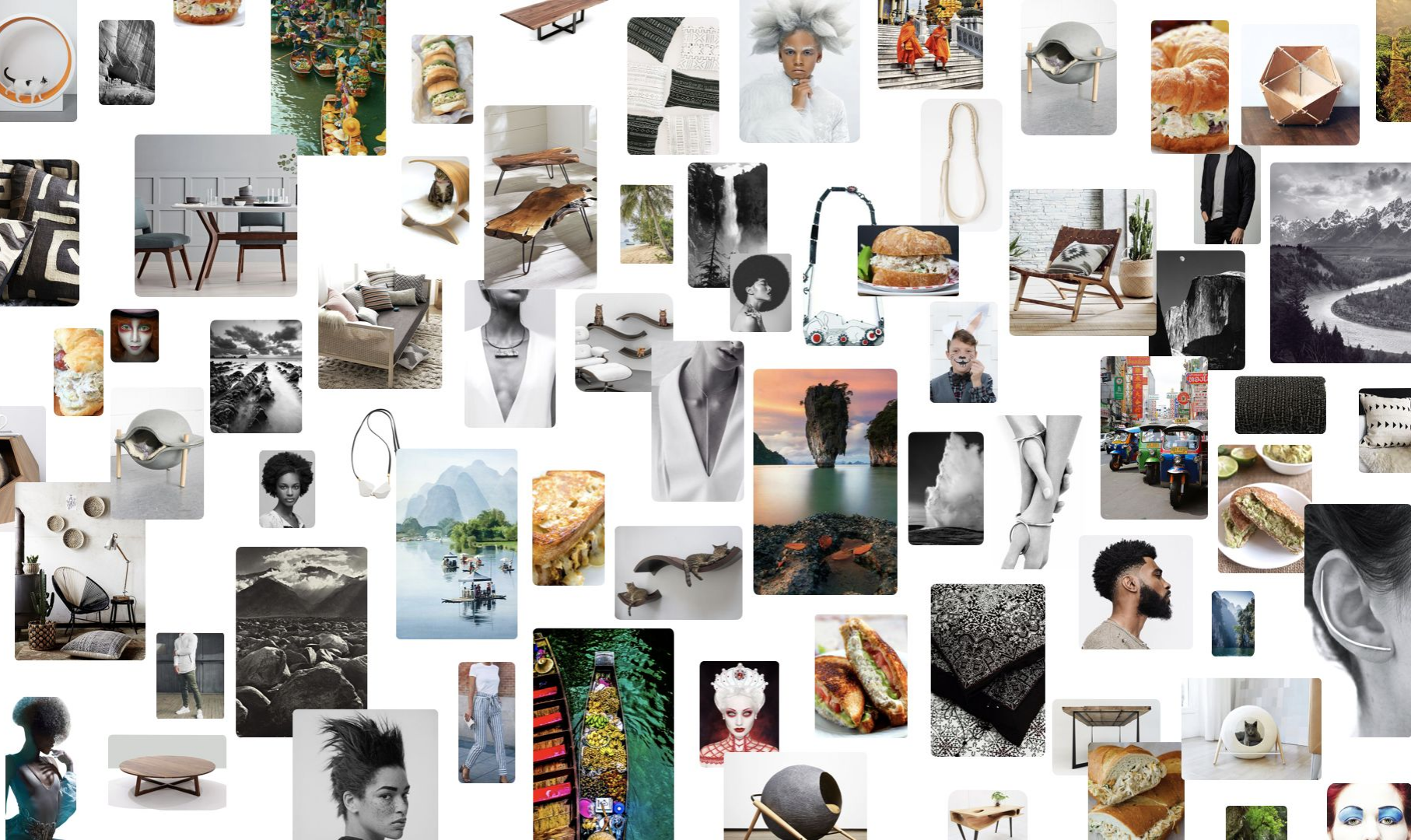
Aug 15th 2021

Introduction

Andrew Zhai

- Senior Staff Applied Scientist
- Deep Learning @ Pinterest, TL of Representation Learning
- Work across recommendation funnel to build scalable ML solutions





Pinterest

Bring *everyone* the *inspiration*
to create a life they love

454 M

Global Monthly Active Users¹

300 B

Pins saved²

6 B+

Boards²

Pinterest is available in more than
30 languages³

91% of Pinners

say Pinterest is a place filled with
positivity⁴

¹ Pinterest, Global analysis, June 2021

² Pinterest, Global analysis, Jan 2021

³ Pinterest internal data, 2020

⁴ Talkshoppe, US, Emotions, Attitudes & Usage study, Oct 2018

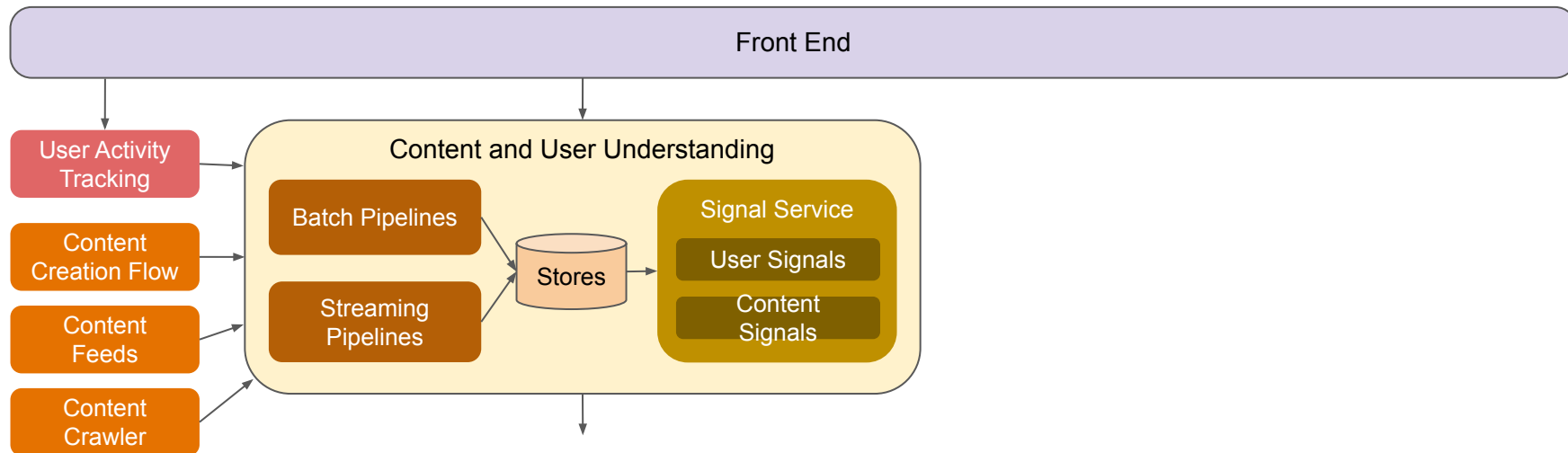


Pinterest Home Feed

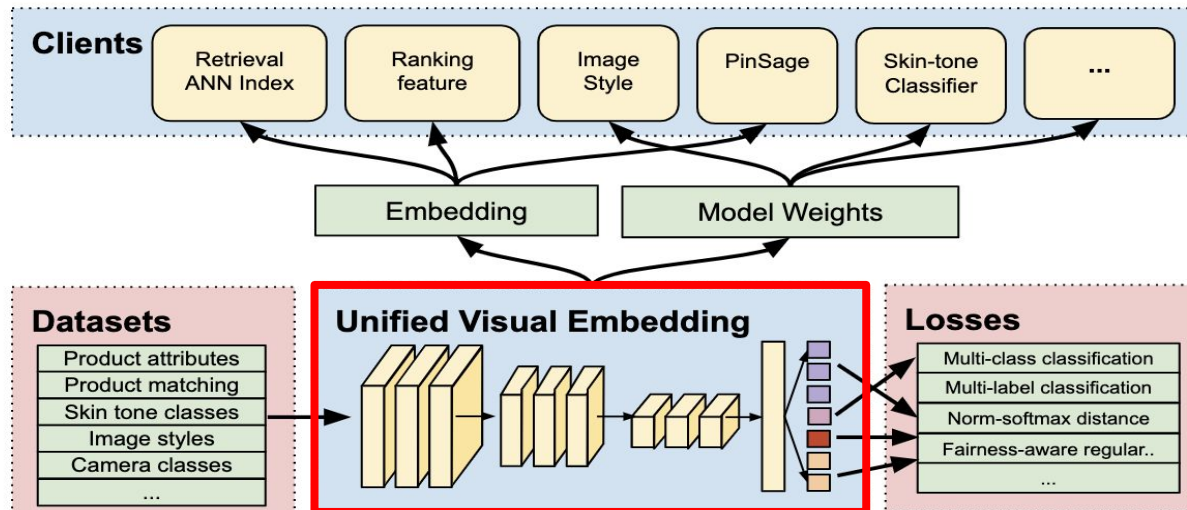
System Architecture for scalability - 00s of thousands of users and few billion Pins (content)

Front End

System Architecture for scalability - 00s of thousands of users and few billion Pins (content)

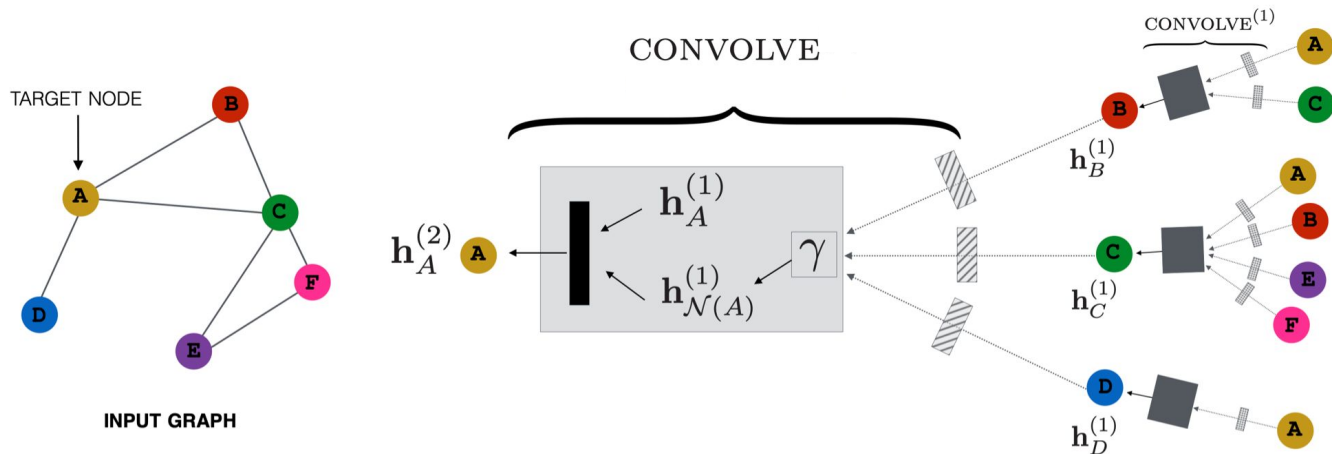


Content Signals: Visual Embeddings



- Input: An image
- Output: An embedding (+ more, later on...)

Content Signals: Graph Embeddings

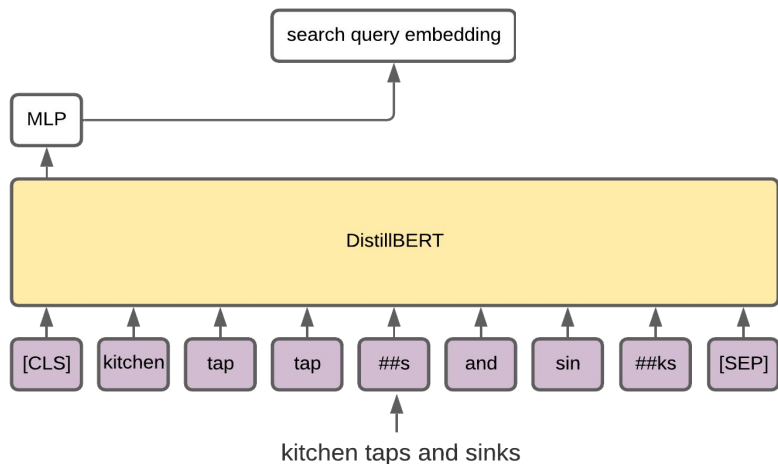


- Combine content and engagement signals in a **inductive** manner to produce more comprehensive representations
- Input: Pin-to-board graph, content features of each node (e.g., text, visual embeddings)
- Output: An embedding per node

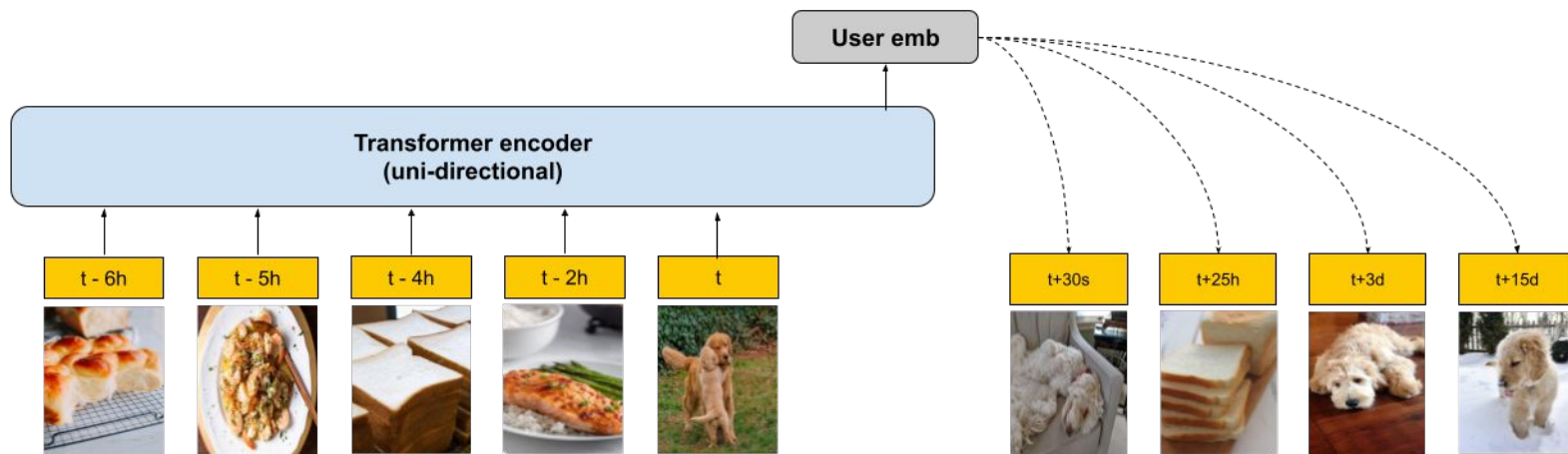
Content Signals: Search Query Embeddings

- Input: Search query (text)
- Output: Embedding optimized for search
- **Tokenized** input for generality

kitchen taps and

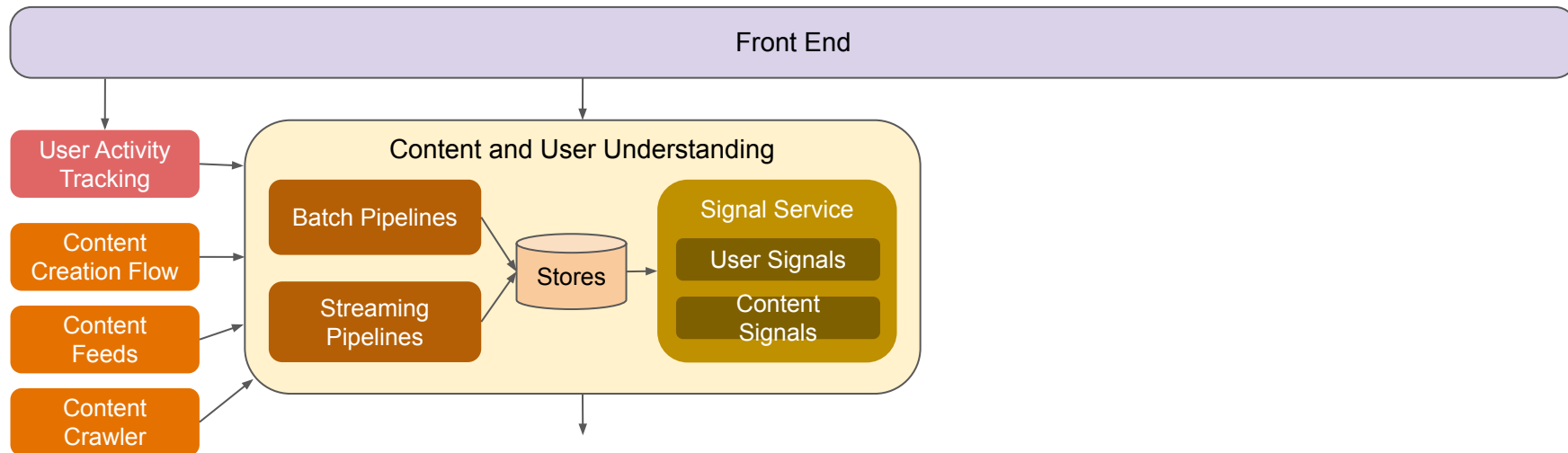


User Signals: User Embeddings

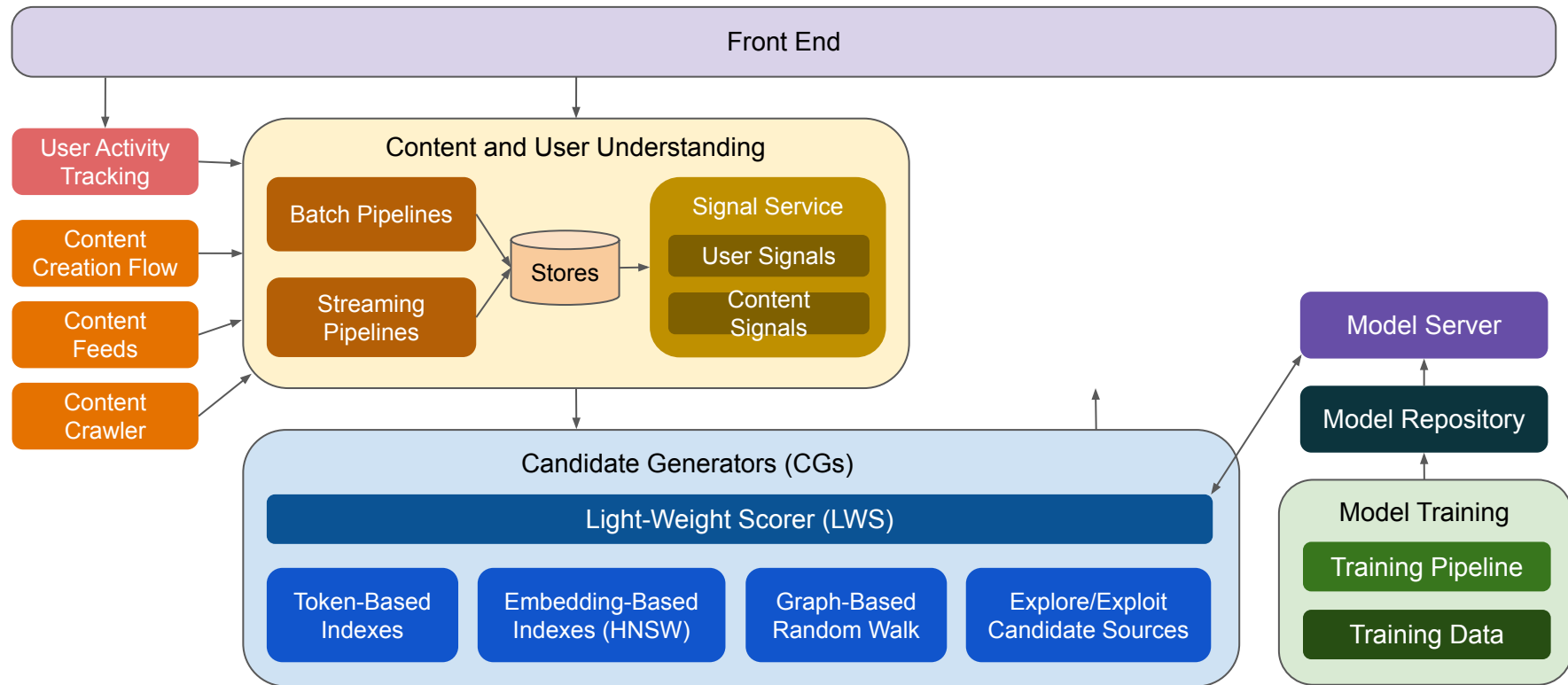


- Input: user activity sequence across all of Pinterest
- Output: one user embedding

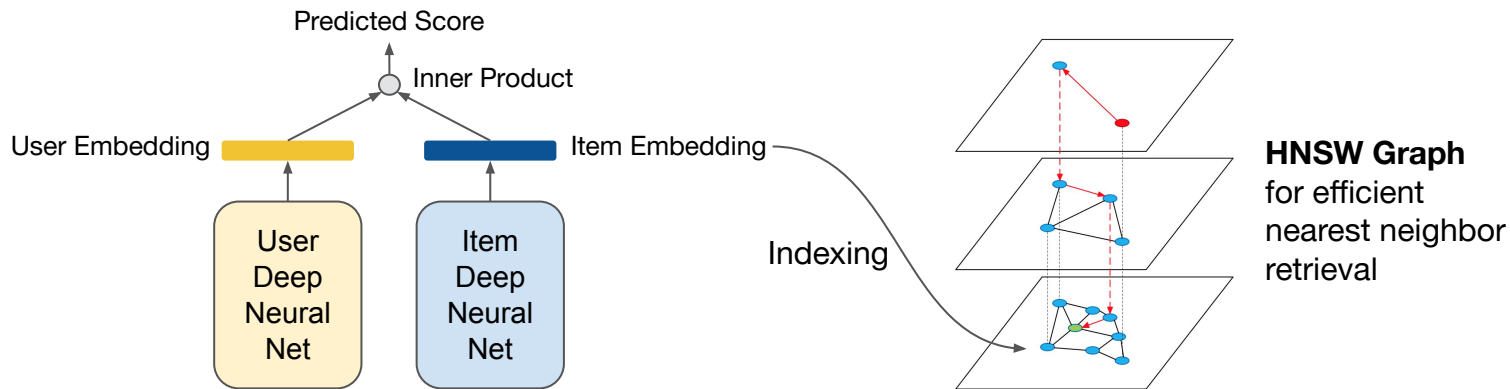
System Architecture for scalability - 00s of thousands of users and few billion Pins (content)



System Architecture for scalability - 00s of thousands of users and few billion Pins (content)

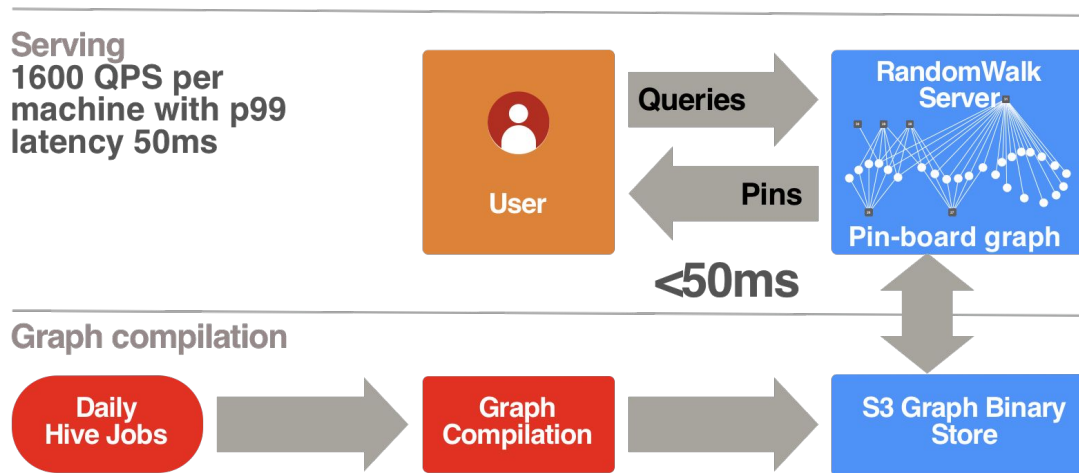


Candidate Generation: Embedding-Based Retrieval



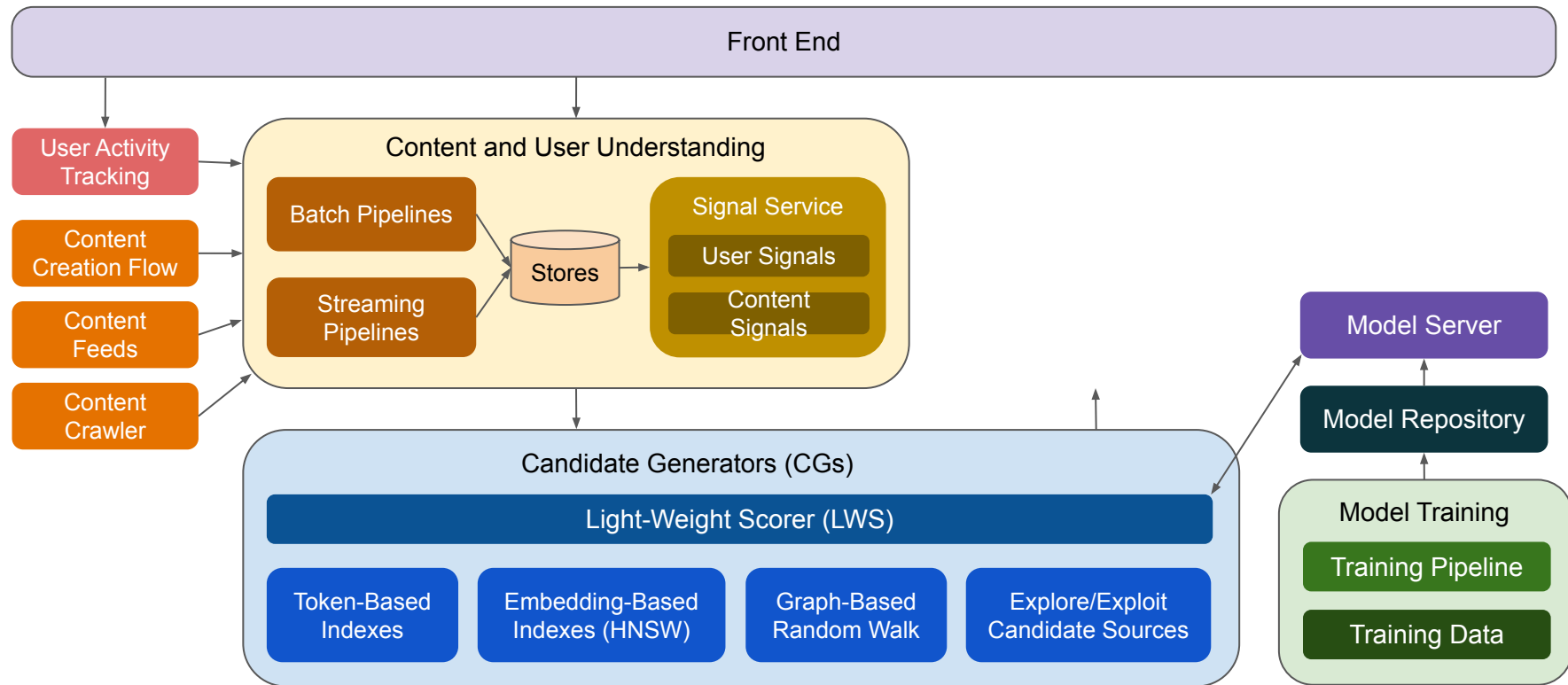
- Train a two-tower deep neural network to predict user engagement
- Precompute the embedding vectors for all items and index them into a Hierarchical Navigable Small World (HNSW) graph
- Given a user embedding vector, retrieve k nearest neighbors (items) based on a learned similarity function (the neural network) through the HNSW graph

Candidate Generation: Random Walk on a Graph

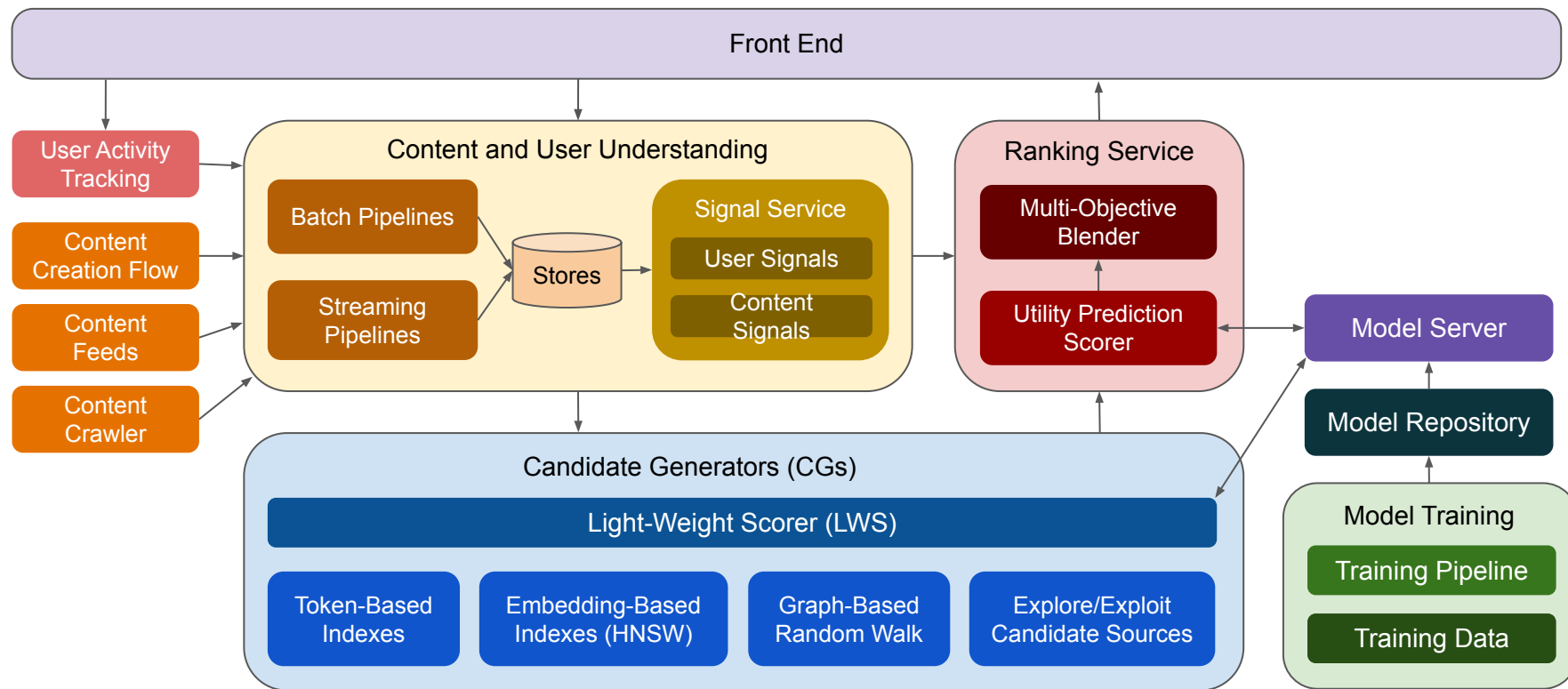


- Start from a set of pins that a user recently interacted with
- Perform random walks from this set of pins
- Return 000s of Pins with the highest visit frequencies (personalized PageRank scores)
- A lot of optimization to make the system highly performant

System Architecture for scalability - 00s of thousands of users and few billion Pins (content)

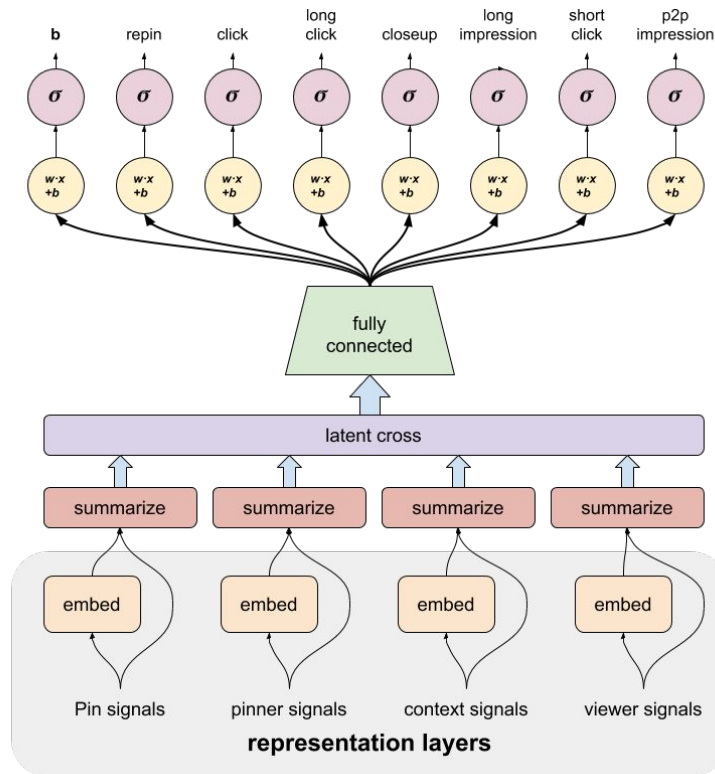


System Architecture for scalability - 00s of thousands of users and few billion Pins (content)



Ranking: User Action Prediction

- Predict a wide variety of user actions for each (user, item) pair through multi-head deep neural network
- User signals include the user's profile, interest vector, and embedding vector of the user's activity sequence (using Transformer)
- Content signals include the item's interest vector, engagement rate estimates, and graph embedding



Ranking: Multi-Objective Optimization

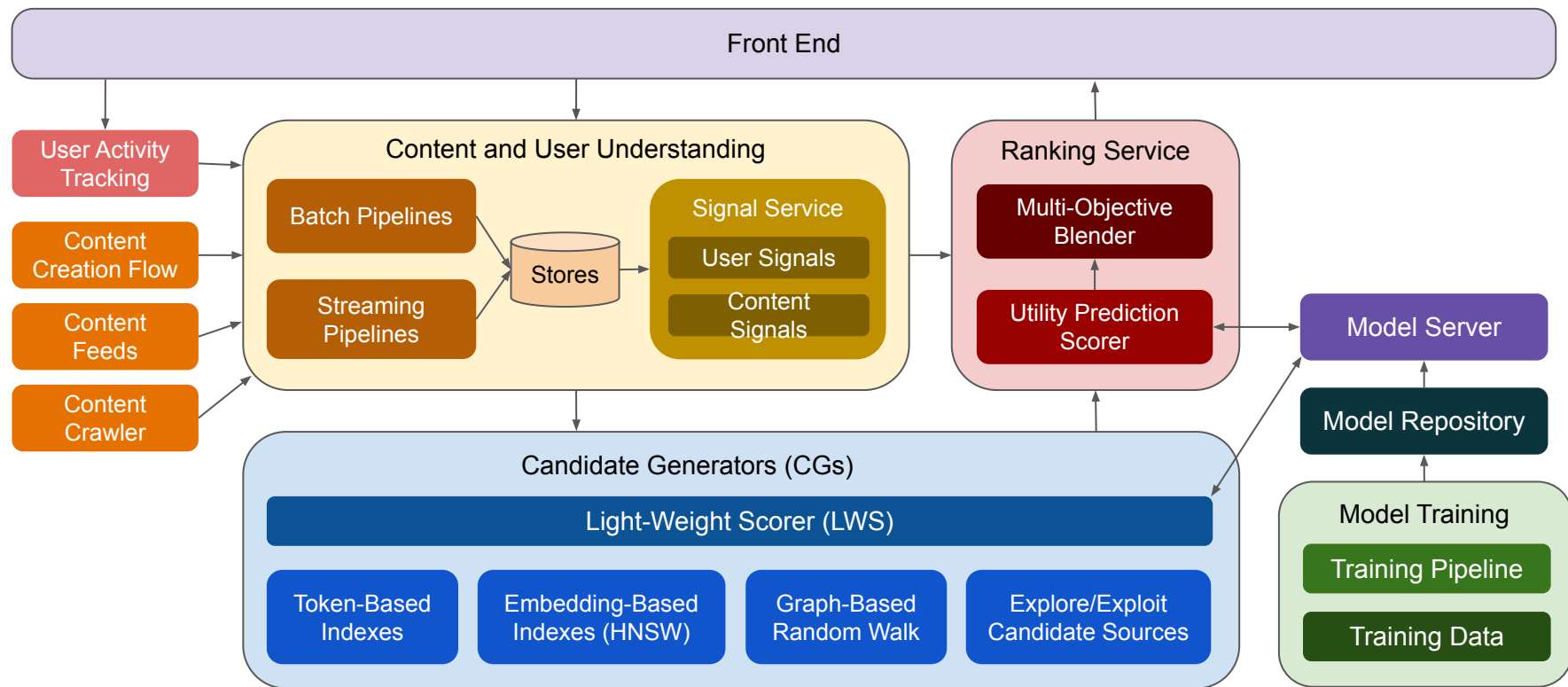
$$\begin{aligned} \max_{\mathbf{x}} \quad & \text{PinnerUtility}(\mathbf{x}) \\ \text{s.t.} \quad & \text{CreatorUtility}(\mathbf{x}) \geq \theta_1 \\ & \text{MerchantUtility}(\mathbf{x}) \geq \theta_2 \\ & \text{AdUtility}(\mathbf{x}) \geq \theta_3 \end{aligned}$$



$$\begin{aligned} \max_{\mathbf{x}} \quad & \text{PinnerUtility}(\mathbf{x}) \\ & + w_1 \text{CreatorUtility}(\mathbf{x}) \\ & + w_2 \text{MerchantUtility}(\mathbf{x}) \\ & + w_3 \text{AdUtility}(\mathbf{x}) \end{aligned}$$

- Estimate utility values for different parties on Pinterest based on predicted action probabilities and causal inference
- Use a simple weighted sum of utility terms
- Tune the weights to achieve a desired tradeoff
- Real system - several non-linearities are present

System Architecture for scalability - 00s of thousands of users and few billion Pins (content)



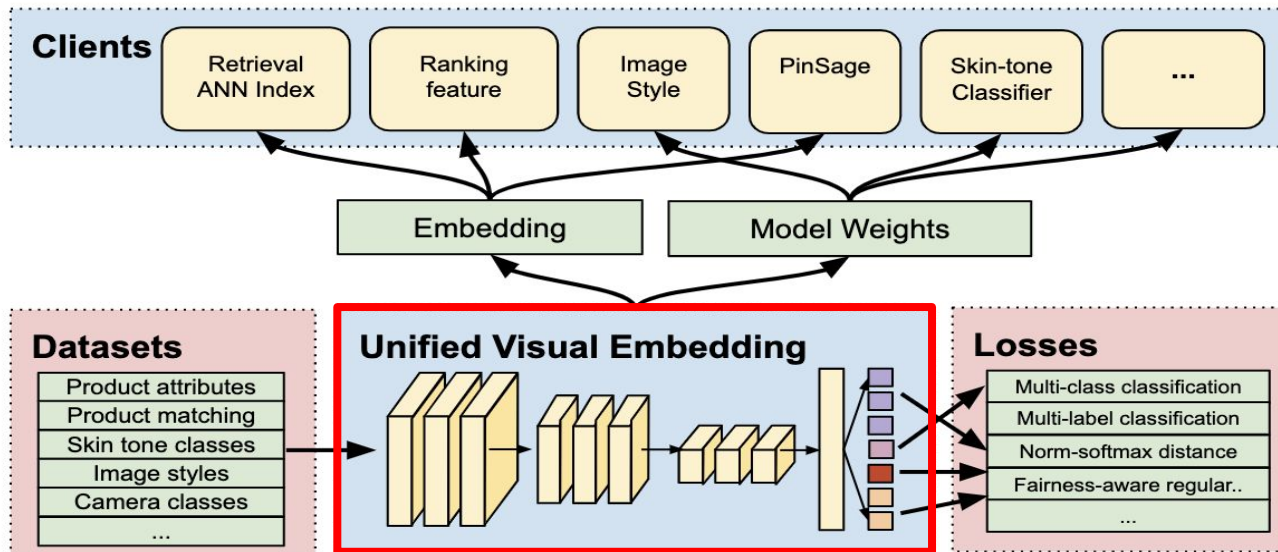


Deep Dive: Representation Learning

Observations

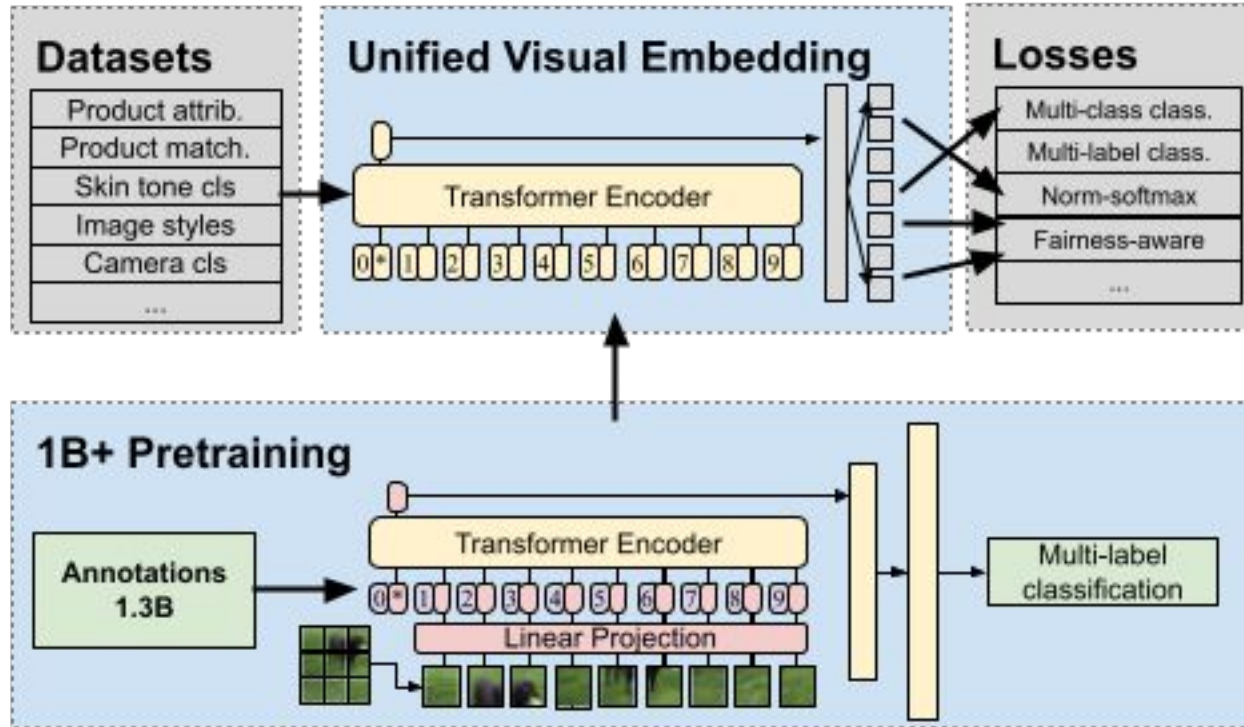
- Request time inference has **tight latency** requirements
 - Ranking scores >10M items per second, p99 < 20ms
 - Need to push complexity offline to signals
- Performance largely depends on how well we understand users, content, search queries, boards:
 - Ranking / retrieval is $f(\text{action} \mid (\text{user}, \text{context}, \text{content}))$
- Homefeed is 1 recommender system, we still have
(search, related pins, board search, ..) x (organic, creator, shopping, advertisement, ..)
 - Need a way to reason about recommender and search systems more uniformly

Visual Embeddings

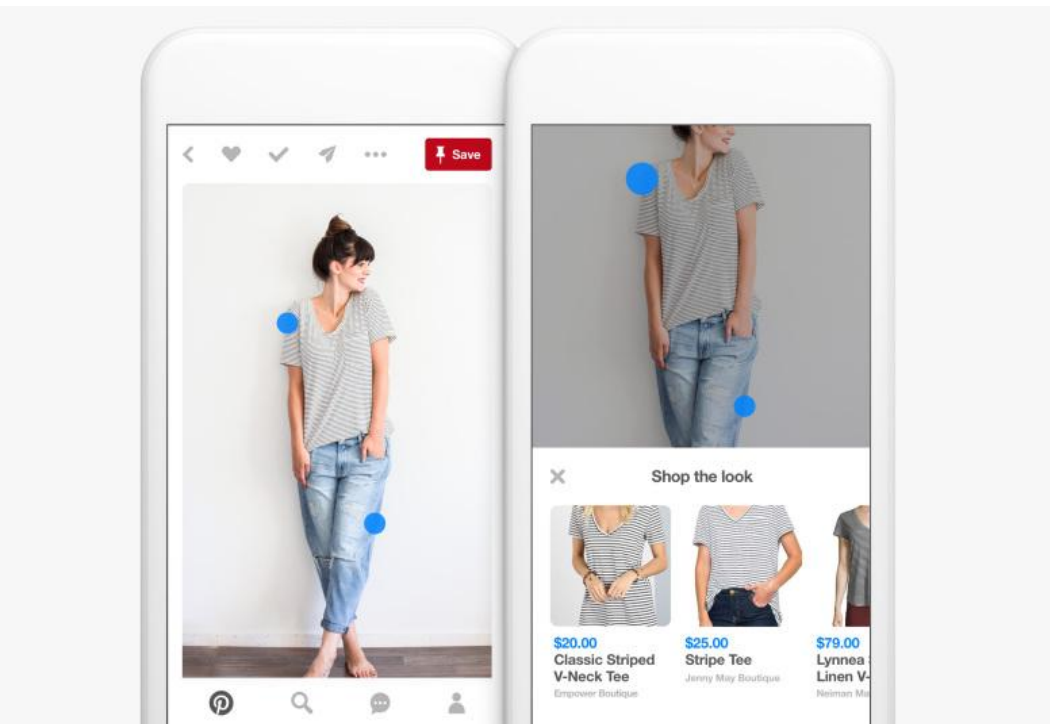


- Input: An image
- Output: embedding, classification, regression, ...
- Multi-task training
 - **15+ objectives** across exact product matching, nearup, skin tone classifier

Pretraining



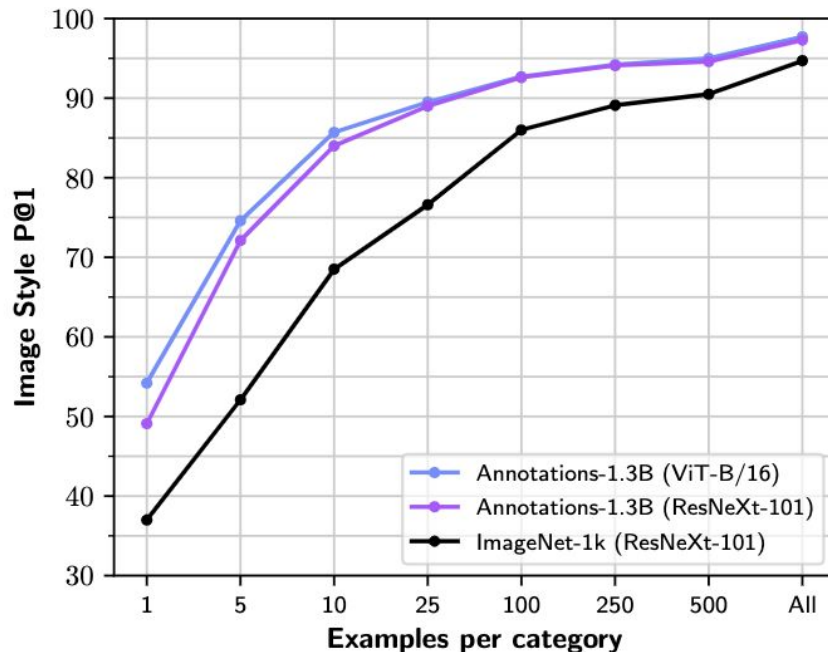
Results



Largest online lifts in Visual Shopping this year

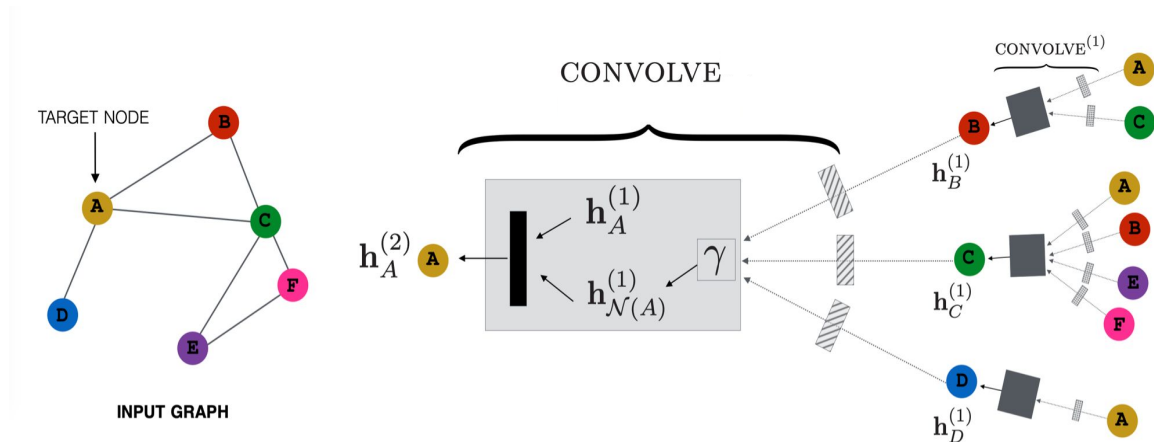
- +38% exact product match@1
- +24-33% long-clickers, and
- +29-30% long click

Few-Shot Learning



Pretraining serves as an effective multiplier on the labeled dataset size.

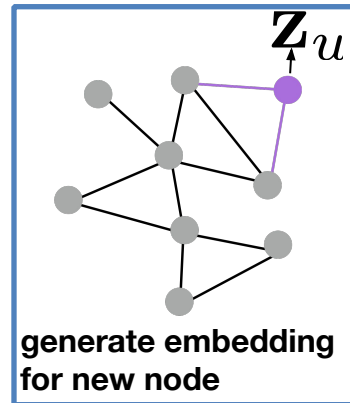
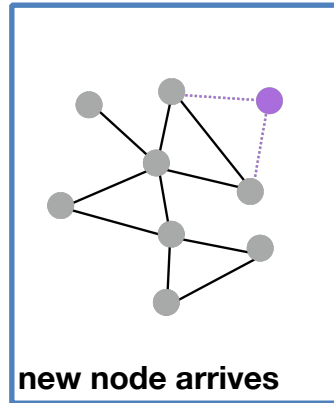
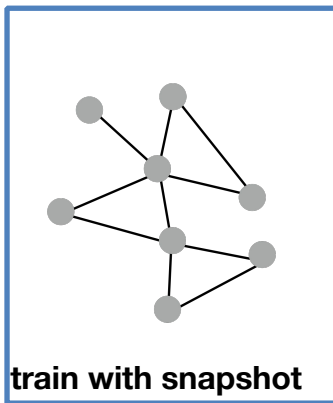
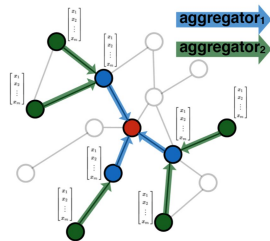
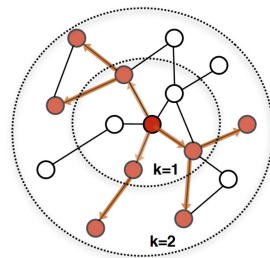
Pin Embeddings (PinSAGE)

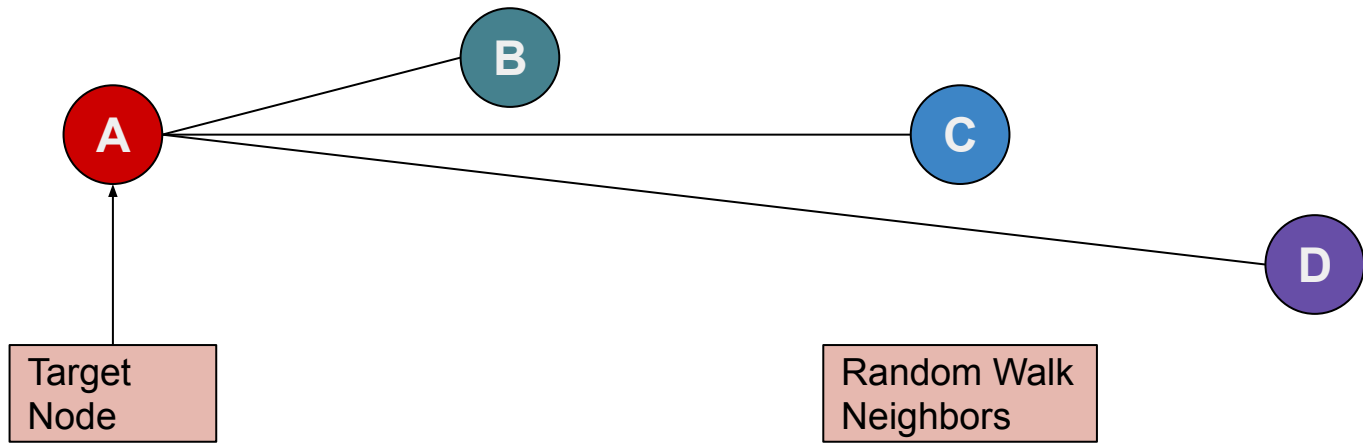


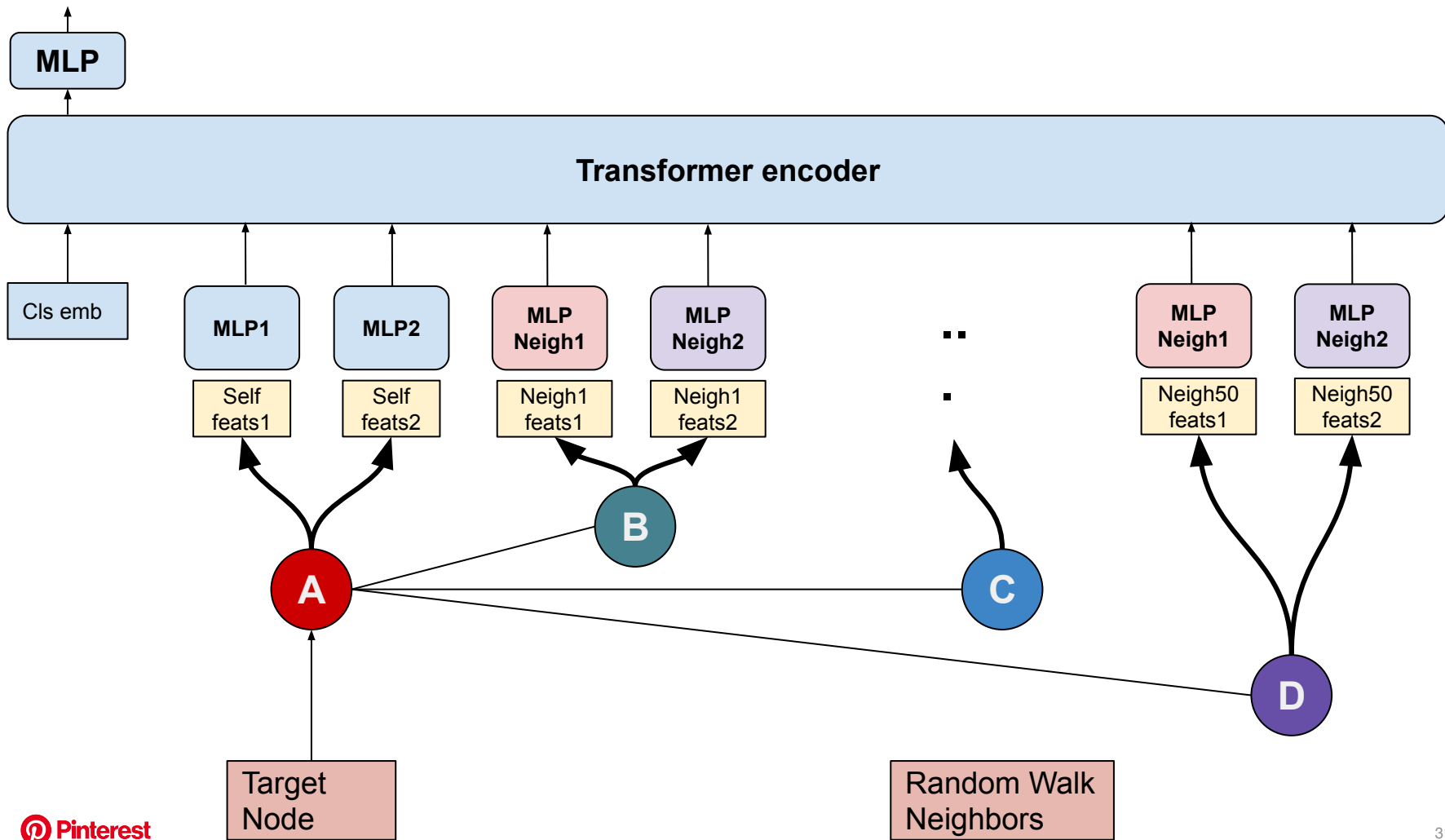
- A **Pin** is predominantly defined by its **content** and **engagement**
 - raw features (visual and text embedding, ...) to represent content
 - pin-board graph to represent engagement
- **>100 use-cases** internally across retrieval, ranking, feature engineering, T&S, diversification for shopping, monetization, creators, etc.

Inductive Learning for Adaptation

- **Dataset:** 3B nodes, 18B edges
- Two Hop Neighborhood Subsampling (V1)
- Random walk-based Neighborhood Subsampling (V2)
 - Approximates personalized PageRank (PPR) score
 - Sampled neighborhood for a node is a list of nodes with the top-K PPR score
- Represent node via context **features** not *unique id* for inductive inference







Softmax for retrieval loss



Query



Positive



Negative

- We leverage a retrieval loss to learn meaningful embeddings
- Softmax to predict (q, p) similarity higher than (q, n) for all $n \in N$
 - Not practical, $|N| > 100M$ in practice
 - Use sampled softmax

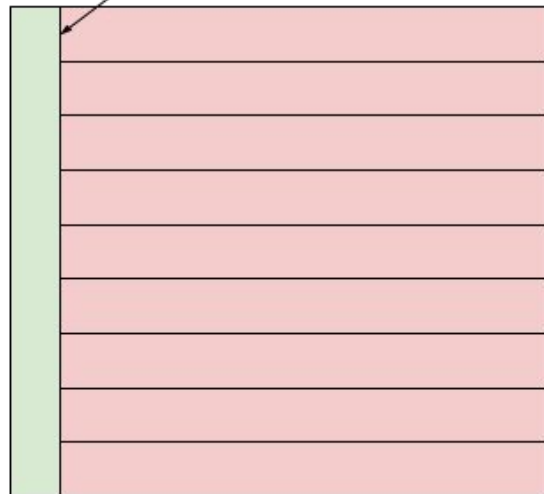
Softmax for retrieval loss

Batch of examples

Query ($N \times d$)
Positive ($N \times d$)
Rand negative ($M \times d$)

Learn to discriminate
(q, p) similarity

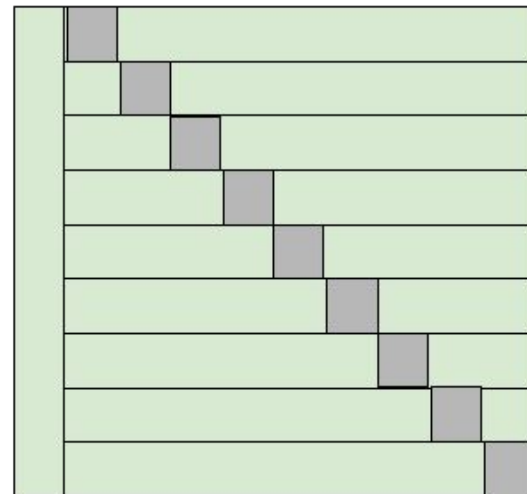
Softmax logits
(rand negatives)



$N \times (1 + M)$

Use **other** positives in
batch as negatives

Softmax logits
(in batch positive as negs)

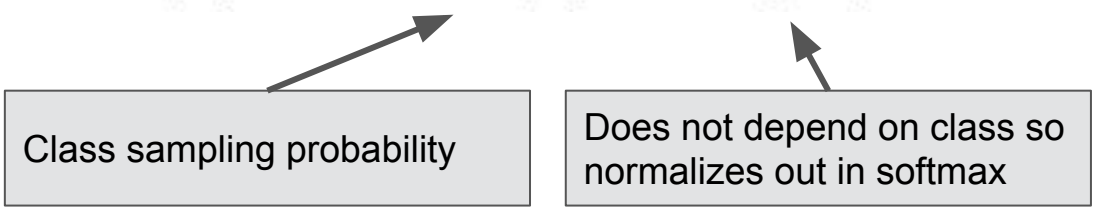


$N \times (1 + N)$

Softmax for retrieval loss

- Probability correction is helpful to remove serving bias
- Sampled softmax logits are predicting $\log(P(y | x_i, C_i))$ (class probabilities over **sampled** classes)
- With some assumptions (e.g. each class is independently sampled), you get:

$$\log P(y | x_i, C_i) = \log P(y | x_i) - \log Q(y | x_i) + K(x_i, C_i)$$



Class sampling probability

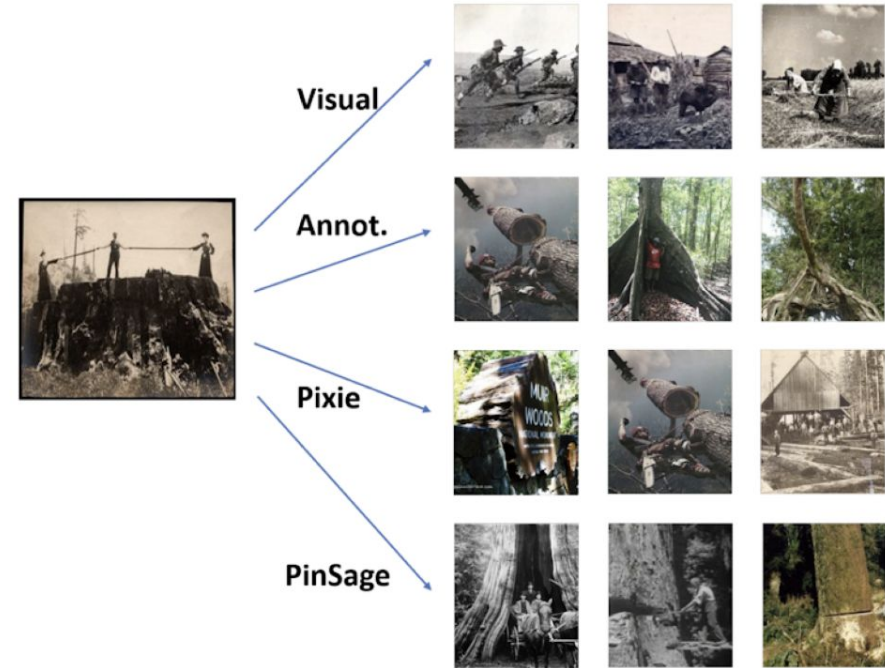
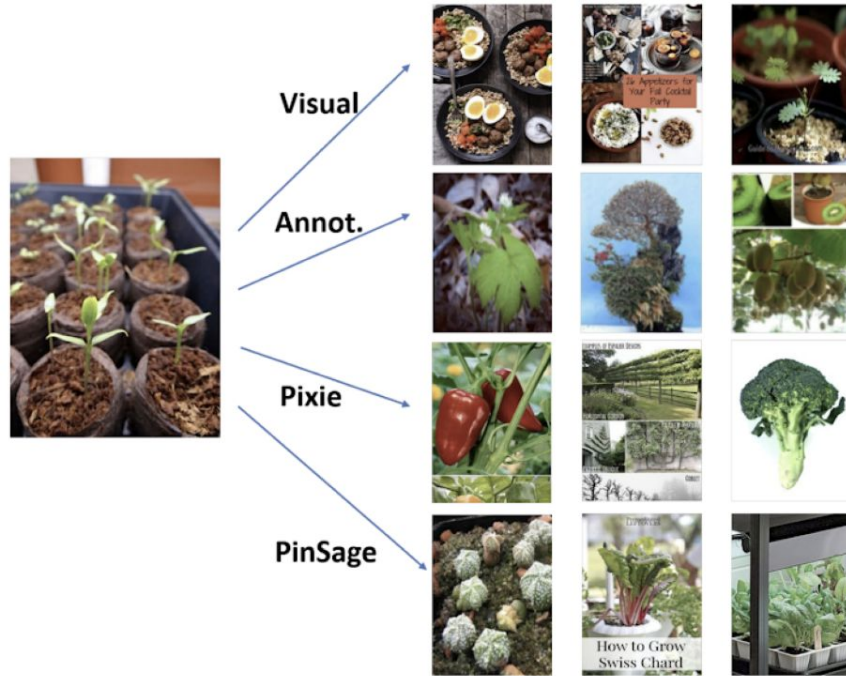
Does not depend on class so
normalizes out in softmax

For a more rigorous treatment:

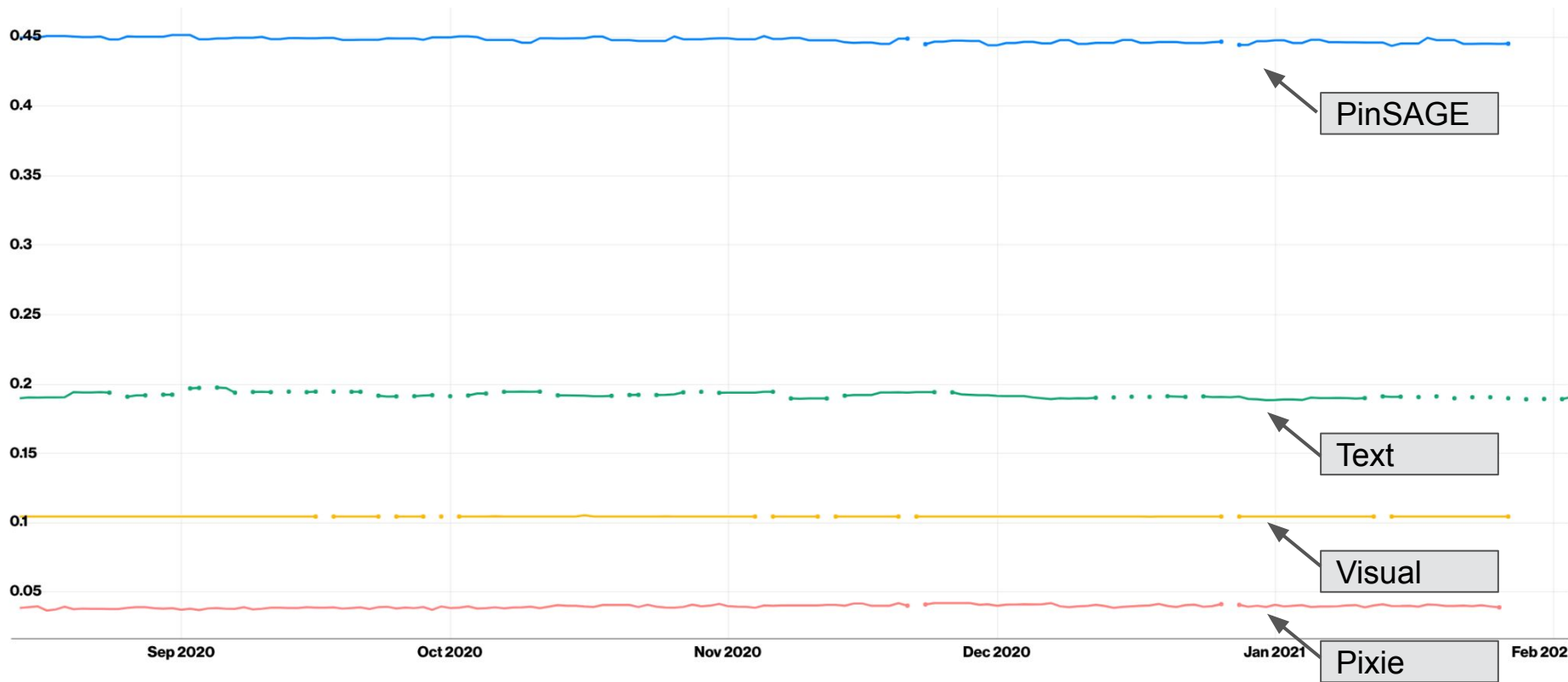
https://www.tensorflow.org/extras/candidate_sampling.pdf

<http://arxiv.org/abs/1412.2007>

Pin Embeddings (PinSAGE)



Recall@1: PinSAGE is performant, and stable

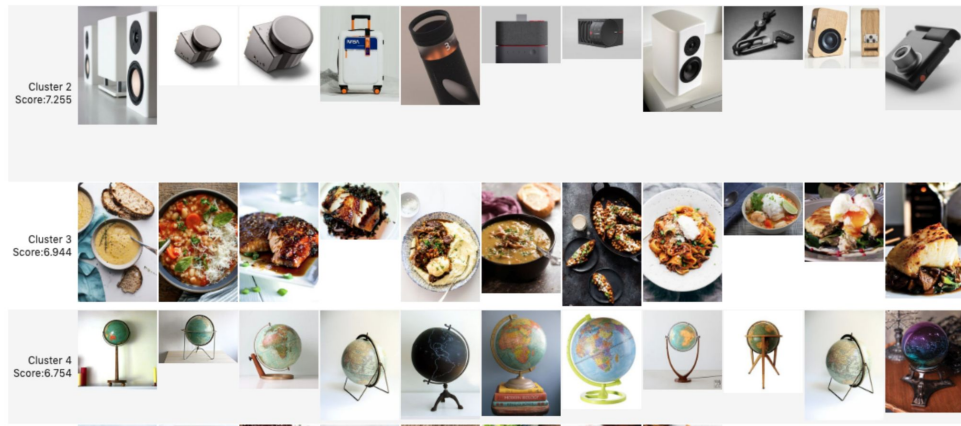


User Signals: User Embeddings

- Our recommender systems already leverage **id** features learned jointly (e.g. in ranking)
- We want a content based signal for users that's more **semantic** and **adapts** online

User Embeddings

PinnerSAGE:
(clustering)



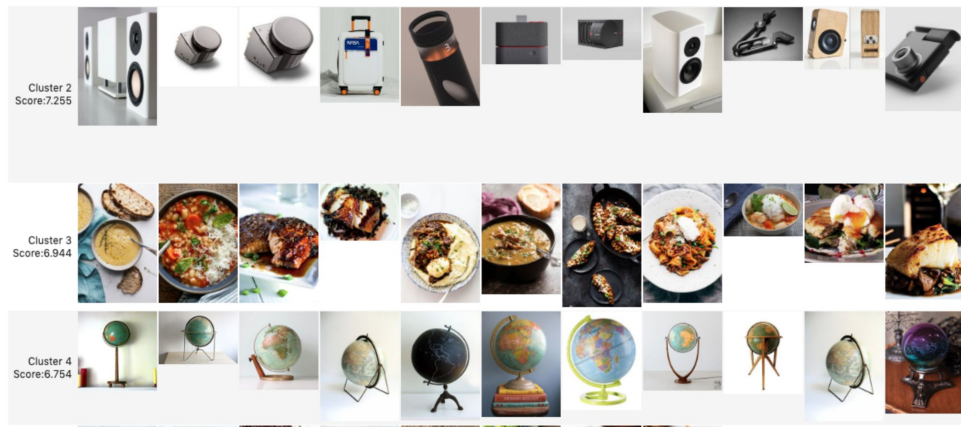
User Emb 1

...

User Emb k

User Embeddings

PinnerSAGE:
(clustering)

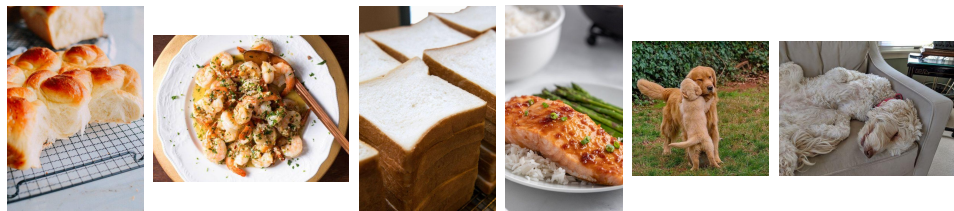


User Emb 1

...

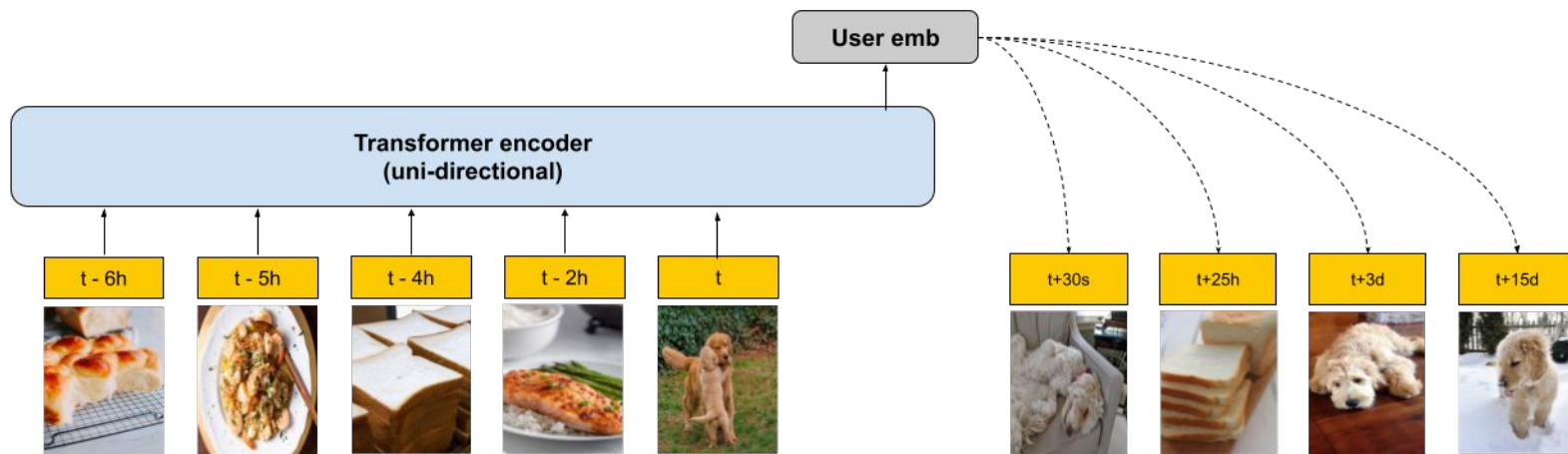
User Emb k

New user
embedding
(Supervised
Transformer)



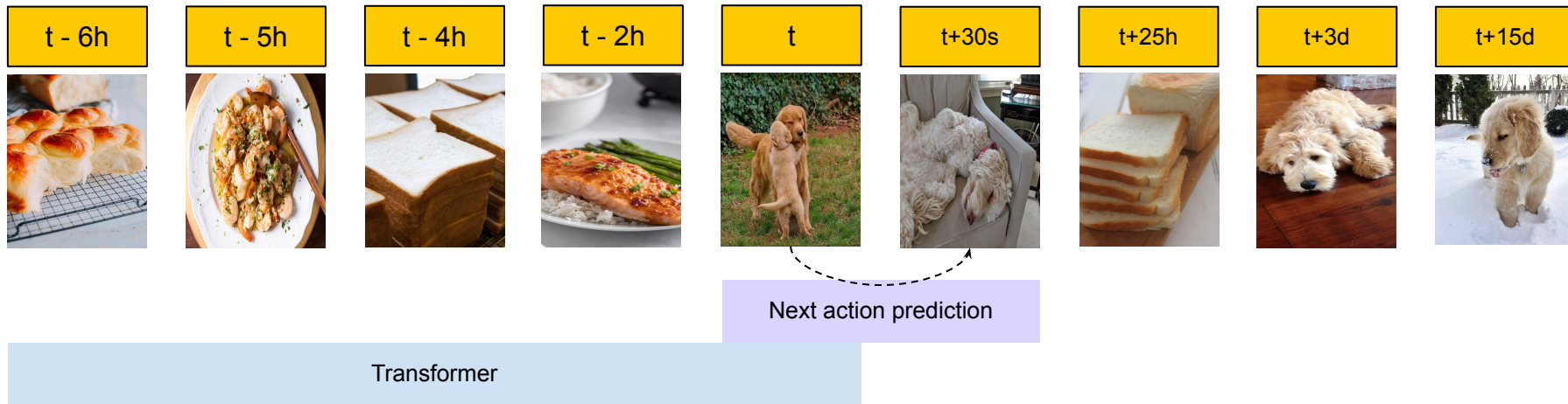
User Emb

User Signals: User Embeddings



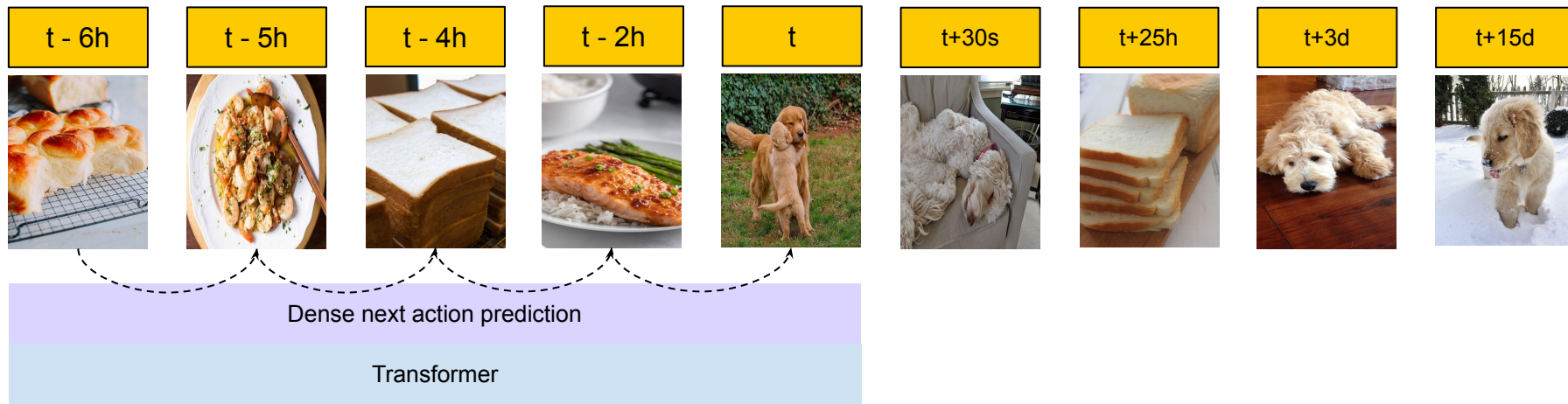
- Input: Last K user activity sequence across all of Pinterest
- Output: one user embedding summarizing activity jointly for short and long-term activity prediction.
- $O(100M \text{ vocab})$ for “action” on item - **softmax retrieval loss** and **PinSage**

Training Objective: Next Action



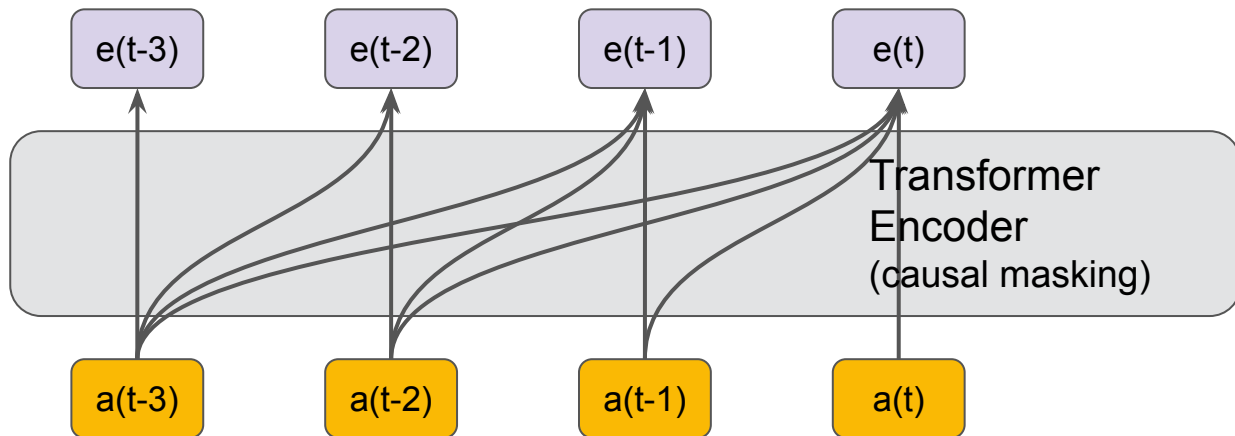
Prediction = one example to softmax retrieval
(user, next action positive, random negatives)

Training Objective: “Dense” next Action

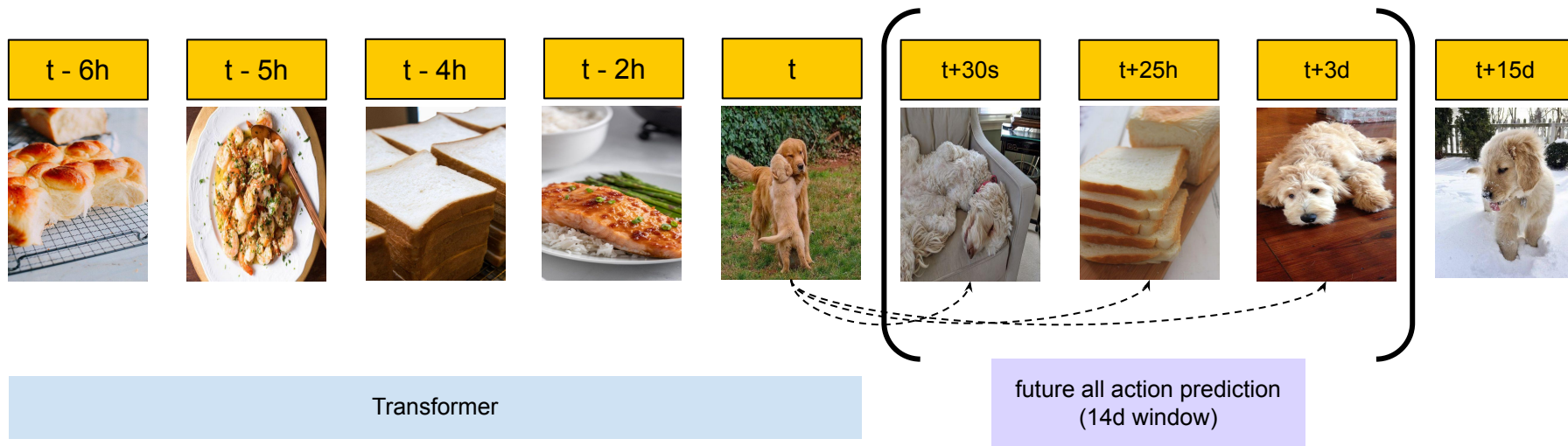


Training Objective: “Dense” Next Action

- $e(t-1)$ predicts $a(t)$
- Only attend to previous actions



Training Objective: All Action

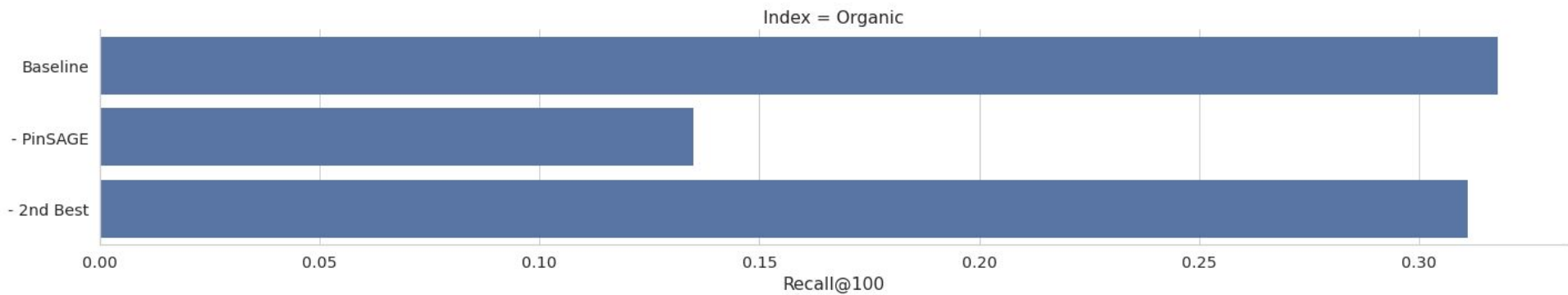


User Embedding Results

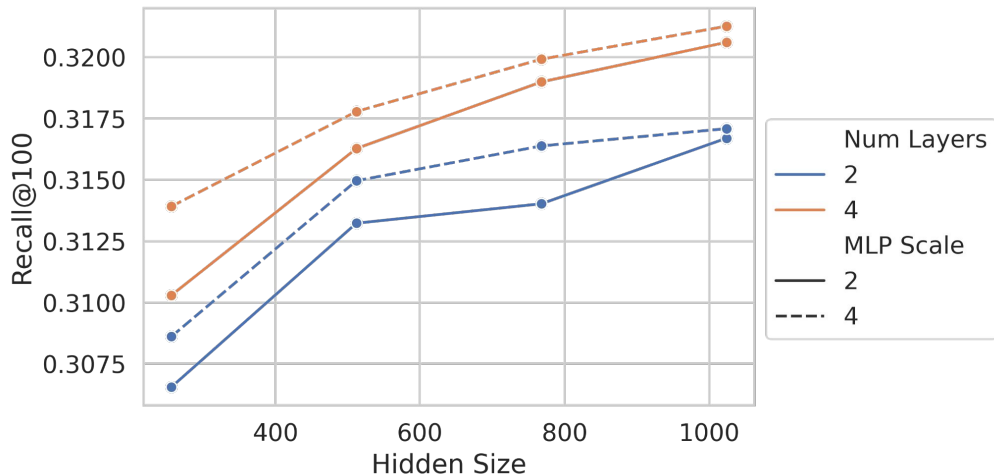
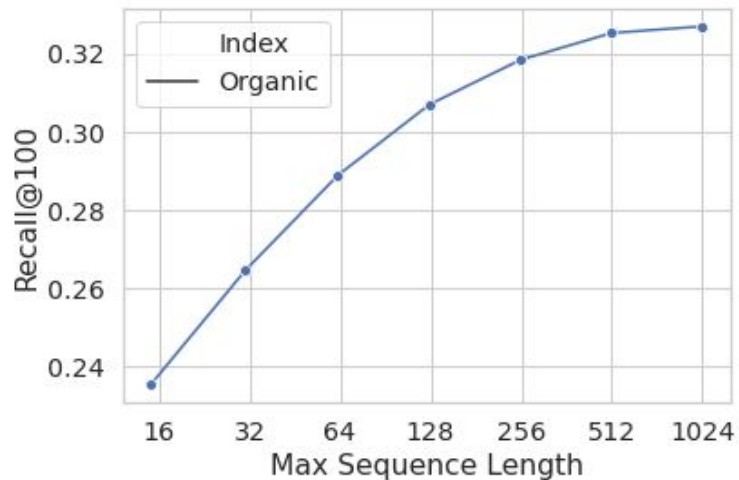
	all_action R@100
(oracle) PinnerSAGE (5 clusters)	0.125
(oracle) PinnerSAGE (20 clusters)	0.205
Our method (1 embedding)	0.255

- Online experiment in Homefeed ranking, replacing prior method
 - **+1-2%** timespent on Pinterest, **+3-4%** engagement lift, **-2.6%** content hides. Wins across in shopping, creators, organic

Ablation: Feature Importance



Ablation: Larger is better



Summary

- Representations are critical to recommendation performance
 - Often times where the most complex ML models reside
- Large capacity models with **huge** data is very useful
- We use **Transformers** everywhere, general feature interaction module
- Build on top of each other (Visual -> PinSage -> PinnerSage)
 - Separate models due to training cost
- Team jointly optimizes **vertically** (users, pin, image, video, text, creator, products, ...) and **horizontally** (softmax retrieval, transformers, ...)



Come join us! <https://www.pinterestcareers.com/>